**101.** A highway bridge uses steel girders.

Design Code:
*AASHTO LRFD Bridge Design Specifications,* 8th edition.

Design Data:

| | |
|---|---|
| Total service dead load moment (DC) | 1,200 ft-kips |
| Total service live load moment including dynamic load allowance (LL + IM) | 850 ft-kips |
| Total service dead load moment due to wearing surface (DW) | 100 ft-kips |

The design moment (ft-kips) using the Strength I loading combination with maximum load factors is most nearly:

(A) 2,912
(B) 3,138
(C) 3,438
(D) 3,684

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**102.** A single-span concrete slab bridge is shown in the figure.

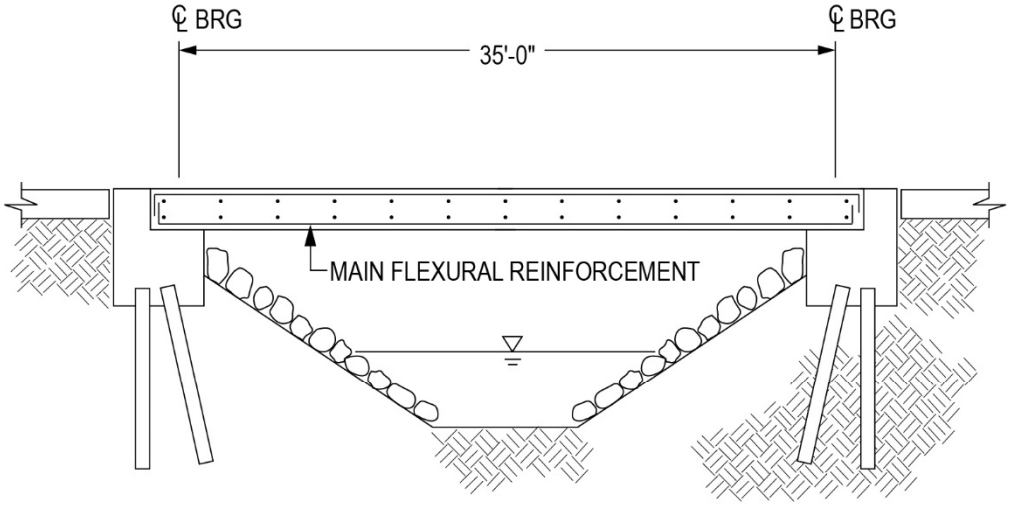
Design Code:
*AASHTO LRFD Bridge Design Specifications*, 8th edition.

Design Data:
Live load HL-93
Unfactored design truck moment per traffic lane 360 ft-kips

Assumption:
Main flexural reinforcement is parallel with traffic.

The maximum factored design vehicular live load moment per traffic lane (ft-kips) for a Strength I limit state, including dynamic load allowance IM, is most nearly:

(A) 630
(B) 838
(C) 1,010
(D) 1,065




ELEVATION

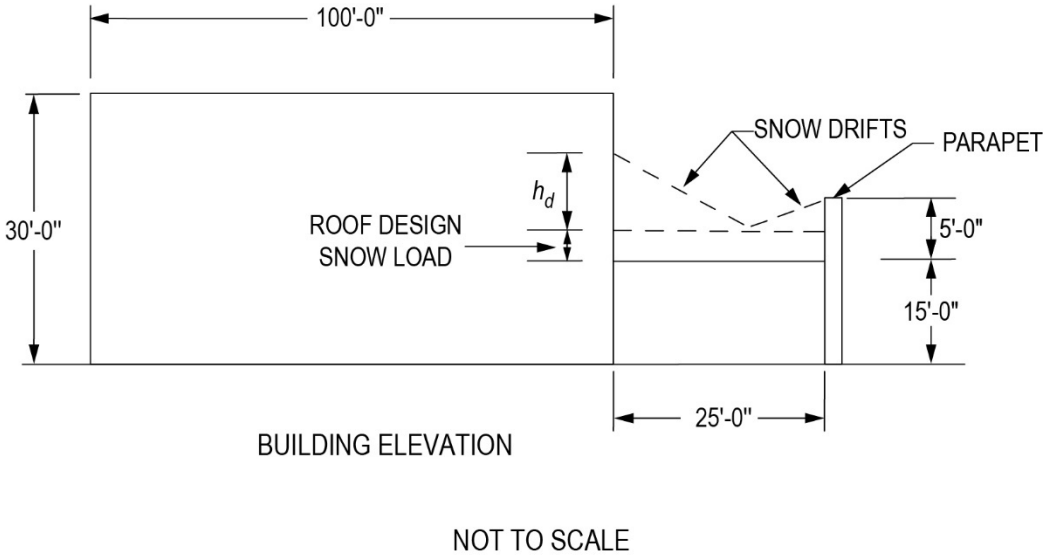
**103.** The figure shows an essential facility that is located in a sheltered area.

Design Codes:
IBC: *International Building Code*, 2018 edition.
ASCE 7: *Minimum Design Loads and Associated Criteria for Buildings and Other Structures*, 2016 edition.

Design Data:
Ground snow load $p_g = 30$ psf

The drift height $h_d$ (ft) shown on the figure is most nearly:

(A) 1.7
(B) 2.5
(C) 3.5
(D) 5.0



**104.** A crane rail supports a 5-kip vertical wheel load from a cab-operated bridge crane.

Design Codes:
IBC: *International Building Code*, 2018 edition.
ASCE 7: *Minimum Design Loads and Associated Criteria for Buildings and Other Structures*, 2016 edition.

The service vertical wheel load (kips), including impact, and the service longitudinal traction load (kips) on the crane rail are most nearly:

| | Vertical Load | Longitudinal Load |
|---|---|---|
| (A) | 4.00 | 1.00 |
| (B) | 5.00 | 5.00 |
| (C) | 6.25 | 0.25 |
| (D) | 6.25 | 0.50 |

**105.** A beam for an exposed roof structure is in St. Louis, MO.

Design Codes:
IBC: *International Building Code*, 2018 edition.
ASCE 7: *Minimum Design Loads and Associated Criteria for Buildings and Other Structures*, 2016.

Design Data:
Beam: W24×55, d = 23.60, $b_f$ = 7.01
Building is an essential facility.
Elevation of beam is 33 ft above grade, $K_{zt}$ = 1.0

The weight of ice (lb) on the beam per linear foot due to freezing rain is most nearly:

(A) 83
(B) 65
(C) 21
(D) 5

**106.** A highway bridge consists of a concrete deck slab supported on steel girders.

Design Code:
*AASHTO LRFD Bridge Design Specifications*, 8th edition.

Design Data:
Live load HL-93
The effective span length is 9'-6".
The deck slab is supported on seven girders.

The negative live load moment (ft-kips) including impact in an interior span of the deck slab per foot width of slab at the centerline of the girder is most nearly:

(A) 6.59
(B) 7.15
(C) 8.76
(D) 9.51

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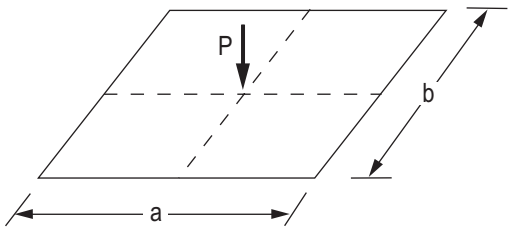
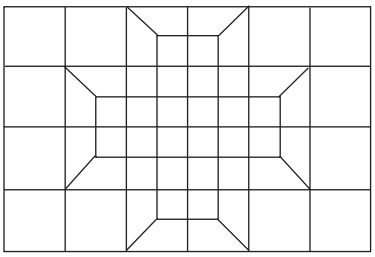
**107.** A plate that is simply supported along its four edges is shown in the figure.

Assumptions:
Use either triangle or quad or combinations of plate elements in the computer model.
Plate elements do not have midside nodes.

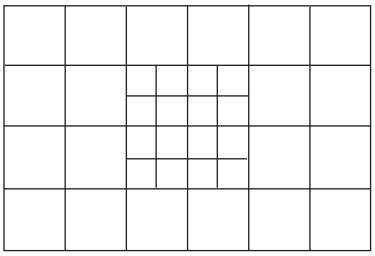
The correct mesh to model the plate for computer analysis is most nearly:



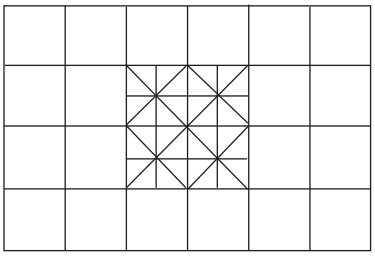

(A)



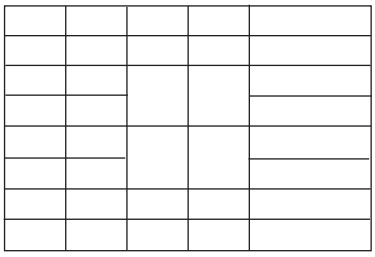
(B)



(C)



(D)



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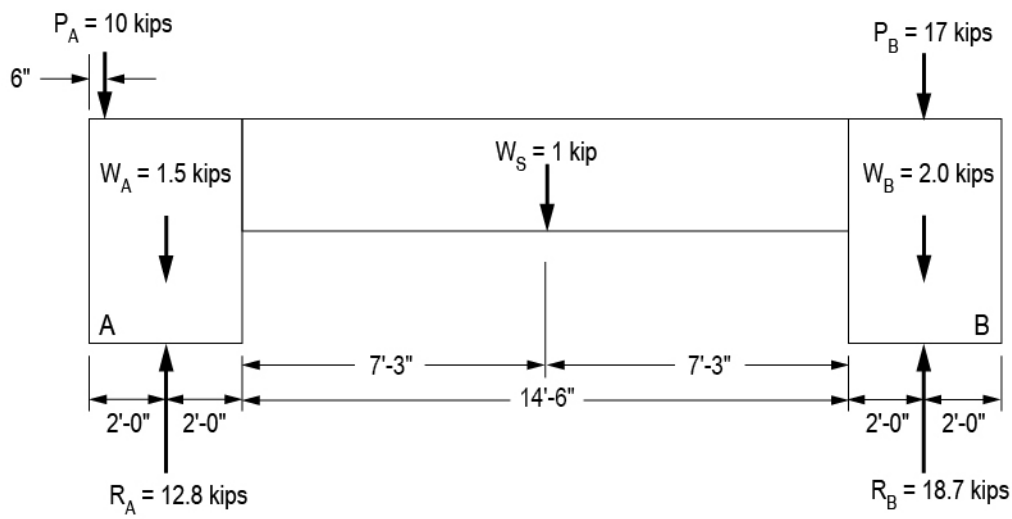
**108.** The eccentrically loaded footing A is connected to the concentrically loaded footing B by a strap beam as shown in the figure.

Design Codes:
IBC: *International Building Code*, 2018 edition.
ACI 318: *Building Code Requirements for Structural Concrete*, 2014 edition.

Assumptions:
The loads and reactions shown have been factored.
Disregard sign convention.

The factored shears (kips) and moments (ft-kips) at the strap beam ends are most nearly:

| | $V_{AB}$ | $M_{AB}$ | $V_{BA}$ | $M_{BA}$ |
|---|---|---|---|---|
| (A) | 12.0 | 38.0 | 19.0 | 38.0 |
| (B) | 13.0 | 25.0 | 20.0 | 40.0 |
| (C) | 1.5 | 30.0 | 0.3 | 0.6 |
| (D) | 1.3 | 12.4 | 0.3 | 0.6 |



**109.** The beam shown is subjected to a uniform load and a moving concentrated load.

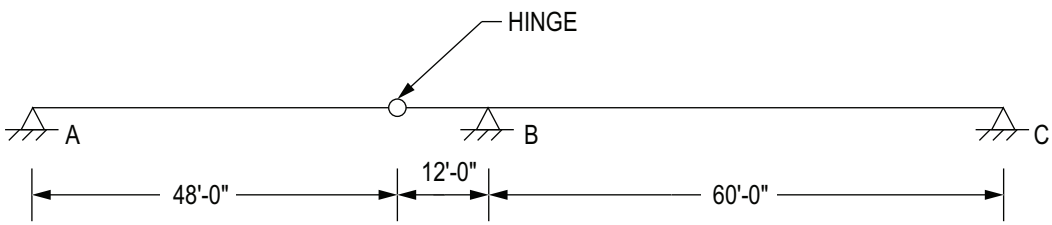
Design Data:

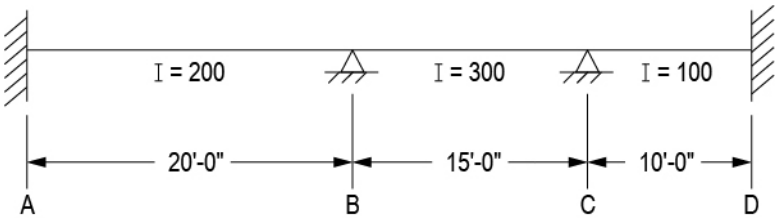
| | |
|---|---|
| Uniform live load | 1 klf |
| Concentrated live load | 10 kips |

Assumption:
The uniform live load may extend the entire length of the beam from A to C.

The maximum live load shear (kips) on the right of Support B is most nearly:

(A) 32
(B) 36
(C) 40
(D) 46




**110.** The figure shows a line diagram for a continuous beam that is supported as shown. The distribution factors at Joint B for Spans BA and BC are most nearly:

| | BA | BC |
|---|---|---|
| (A) | 0.33 | 0.67 |
| (B) | 0.43 | 0.57 |
| (C) | 0.50 | 0.50 |
| (D) | 0.67 | 0.33 |